

Biodiversity-Change Observation With Machine Learning

Thesis Presentation

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Background

- Understanding and Tracking changes in biodiversity is crucial for maintaining a healthy environment. However, this can be a complex task due to the sheer number of species worldwide.
- "Bienenroute" aims to strengthen, observe, and evaluate native floral diversity by encouraging individuals to create diverse flowering spaces for insects, thus promoting biodiversity and species diversity.
- "Bienenroute" allows individuals to directly contribute to biodiversity conservation and learn about the importance of pollinators, demonstrating how individual actions can collectively make a significant impact

Sichtungen
Trachtpflanzen
Anmelden



Die Bienenroute

Die bunte Blütenvielfalt stärken, beobachten und

Blütenvielfalt schafft Artenvielfalt

kann ein bunter Lebensraum werden. Jede Blüte zählt, um das Nahrungsangebot an Pollen und Nektar für Bienen, Hummeln & Co beim Essen und Trinken. Erlebe Szenen wie im Paradies.

Aim

Our main aim is to harness the power of Machine Learning to identify and track changes in plant and insect species within a specific region using an image classification algorithm and deploy in our website (**Bienenroute**).

Objective

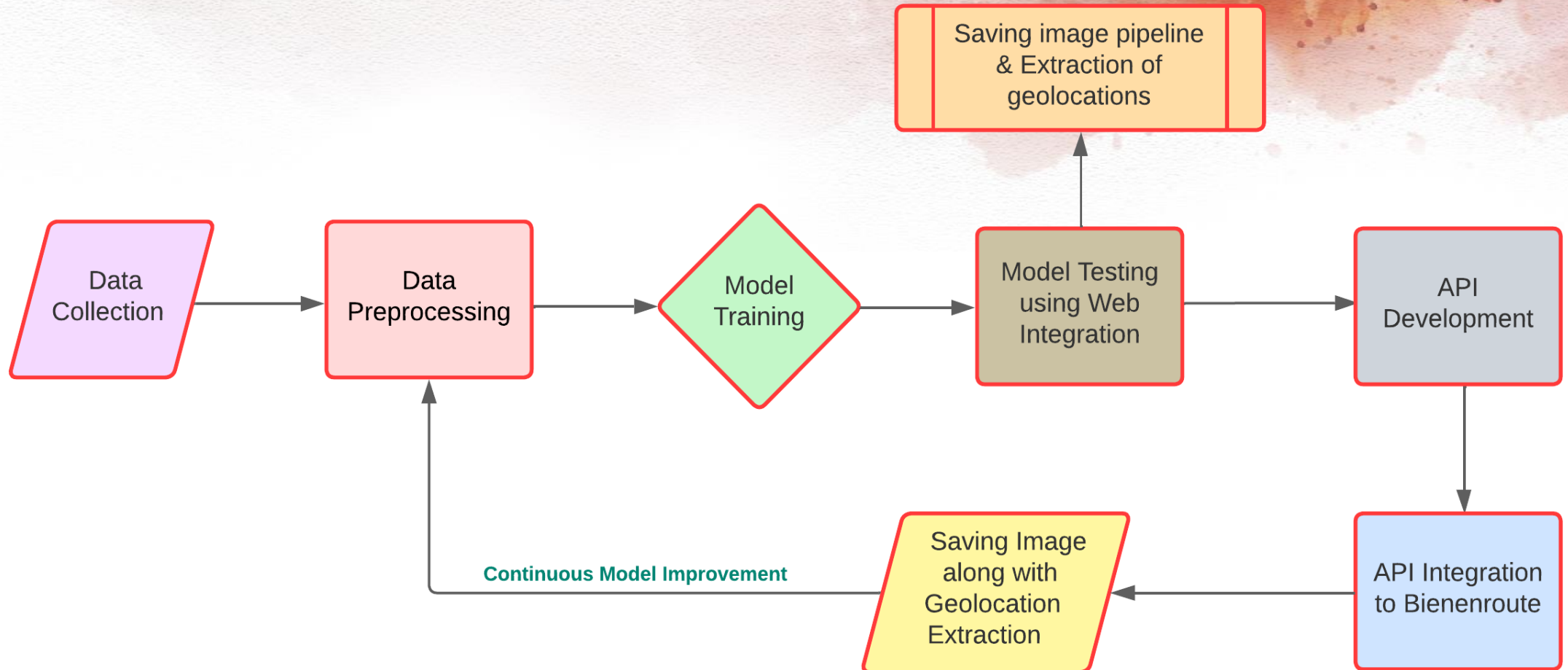
Create a machine learning model to identify and differentiate species from collected images.

Extract geo-location data from the uploaded image to track the spatial distribution.

Development of API integrated Machine Learning Model to "*Bienenroute*"

Enhance model accuracy through Continuous Training.

Process Flow



Data Collection

To source the necessary data for training the image classification model.

Images Sourced:

Species	No of Image	No of Species
Flowers	300 thousand	300
Insects	12 thousand	7

Data Source:

- [PlantNet](#)
- [iNaturalist](#)

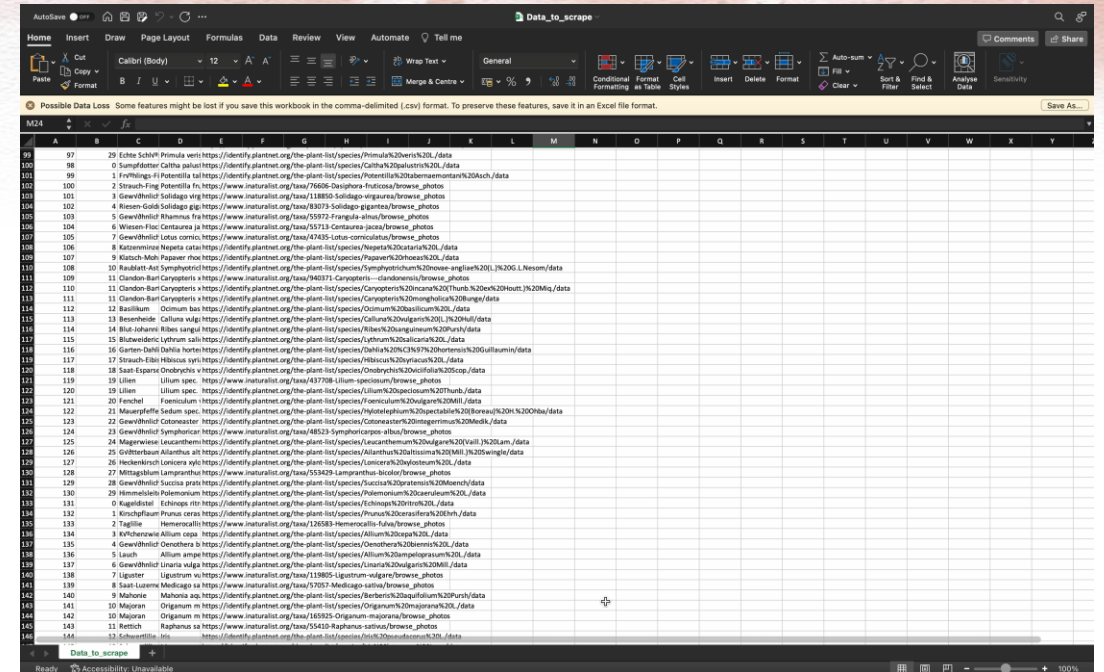
Reasons for Choosing these two websites:

- Already Cover Wide Range of Species
- Support for Open Data and Research
- High Quality Images

Tool Used:

Web-Scraping with Selenium:

An automated tool used for interacting with web browsers, simulating actions like clicking and scrolling.



Data Pre-Processing

Pre-Processing:

Pre-processing prepares the data for the model. It involves steps like:

- **Rescaling:** Adjusting pixel values to be between 0 and 1.
- **Splitting:** Dividing data into training and validation sets.
- **Resizing:** Making all images the same size.

Augmentation:

Augmentation makes the data more diverse. It includes steps:

- **Rotation:** Turning images by a certain degree.
- **Shifting:** Moving images up, down, left, or right.
- **Zooming:** Making images look closer or farther away.



Fig: Example Image with data augmentation

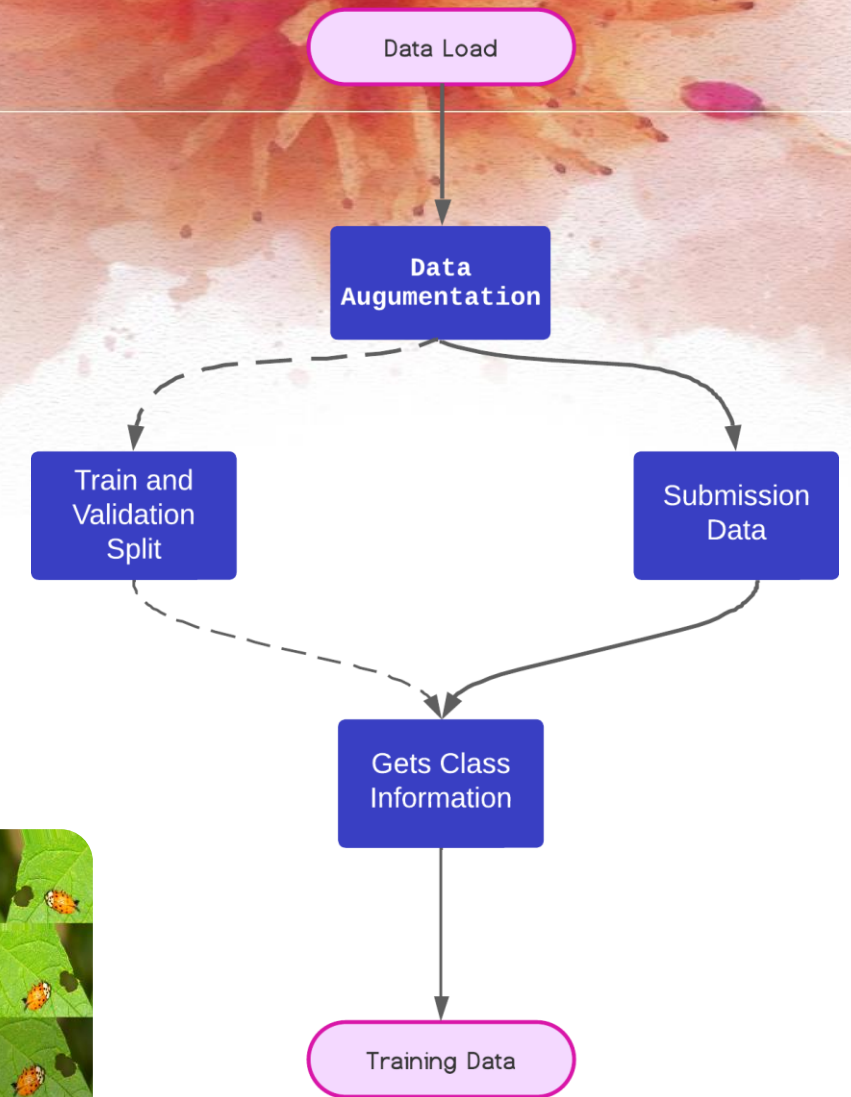


Fig: Data Pre Processing Pipeline

Model Training

Choice of Architecture:

Convolutional Neural Networks (CNN)

Why a CNN Architecture:

- Exceptional performance in image classification tasks
- specifically designed to recognize visual patterns in pixel images, making them ideal for our project involving plant and insect.
- CNNs excel in automatically learning features from images, which eliminates the need for manual feature extraction.

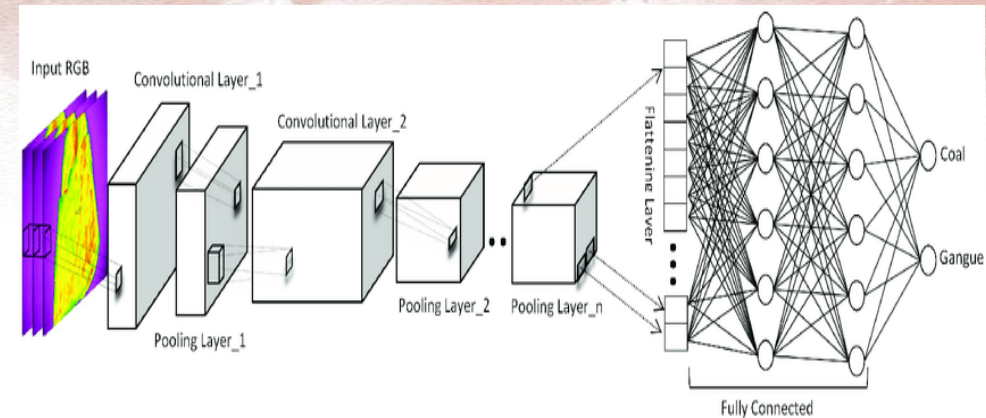
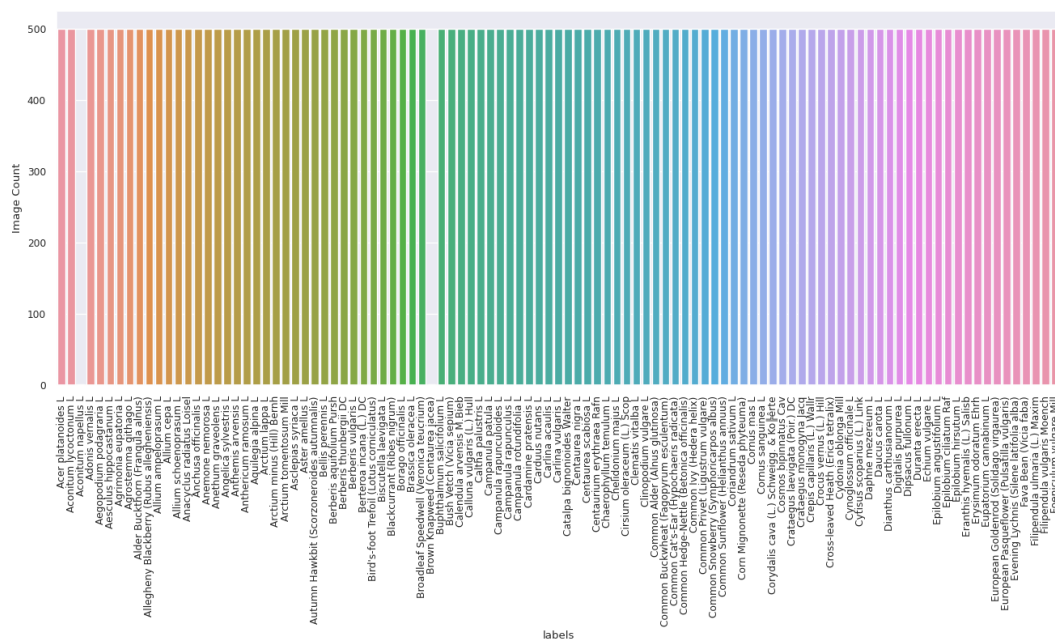


Fig: Construction of the CNN model

Chosen Model For Training:

Model Choice	Reason
Custom Model	A simple model designed for this particular task.
ResNet50	Deep model effective for complex tasks due to its 'skip connections'.
Inception (ImageNet)	Efficient model from Google can also run in low configuration systems.
DenseNet	Connects all layers together for better feature use. Known for top performance.

Model 1 - Custom Model
Model 2 - Pretrained Model using ResNet 50
Model 3 - Pretrained Model using ImageNet
Model 4 - Pretrained Model using DenseNet



Best validation accuracy achieved in each model

Y-axis: Best Val Accuracy (0.1 to 0.7)

X-axis: Epochs (0 to 150)

Legend:

- Model 1 (Blue line)
- Model 2 (Orange line)
- Model 3 (Green line)
- Model 4 (Red line)

Epochs	Model 1	Model 2	Model 3	Model 4
0	0.05	0.05	0.05	0.05
20	0.48	0.45	0.50	0.45
40	0.58	0.55	0.60	0.58
60	0.63	0.60	0.65	0.65
80	0.66	0.63	0.68	0.68
100	0.68	0.65	0.70	0.71
120	0.70	0.67	0.72	0.72
140	0.71	0.68	0.73	0.73
150	0.51	0.48	0.68	0.73

Fig: Validation Accuracy Trend Across Epochs for Multiple Models

Model Training Results For Insect Classification

Model trained with 7 classes Insects with 9000 Images in total

Model 1 - Custom Model

Model 2 - Pretrained Model using ResNet 50

Model 3 - Pretrained Model using ImageNet

Model 4 - Pretrained Model using DenseNet

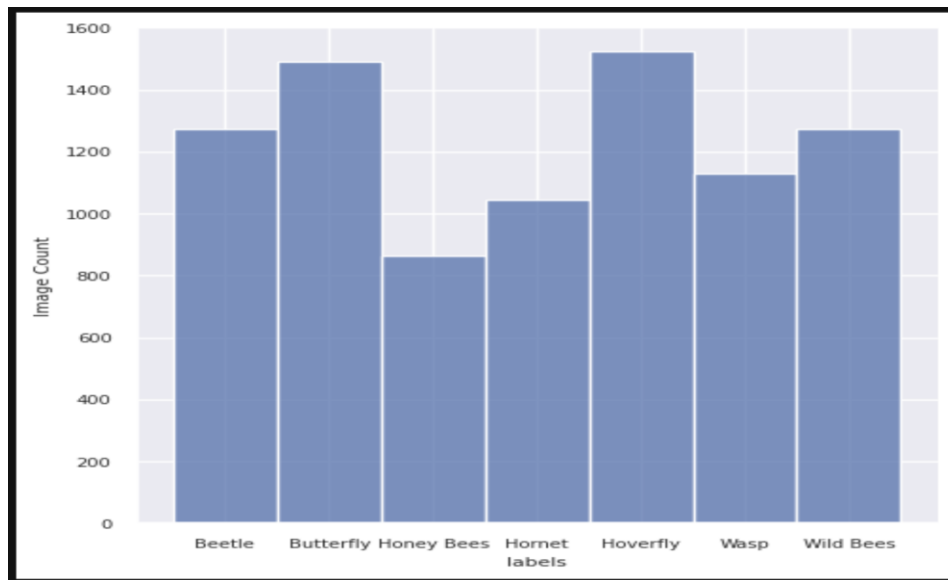


Fig: Data Distribution Plot

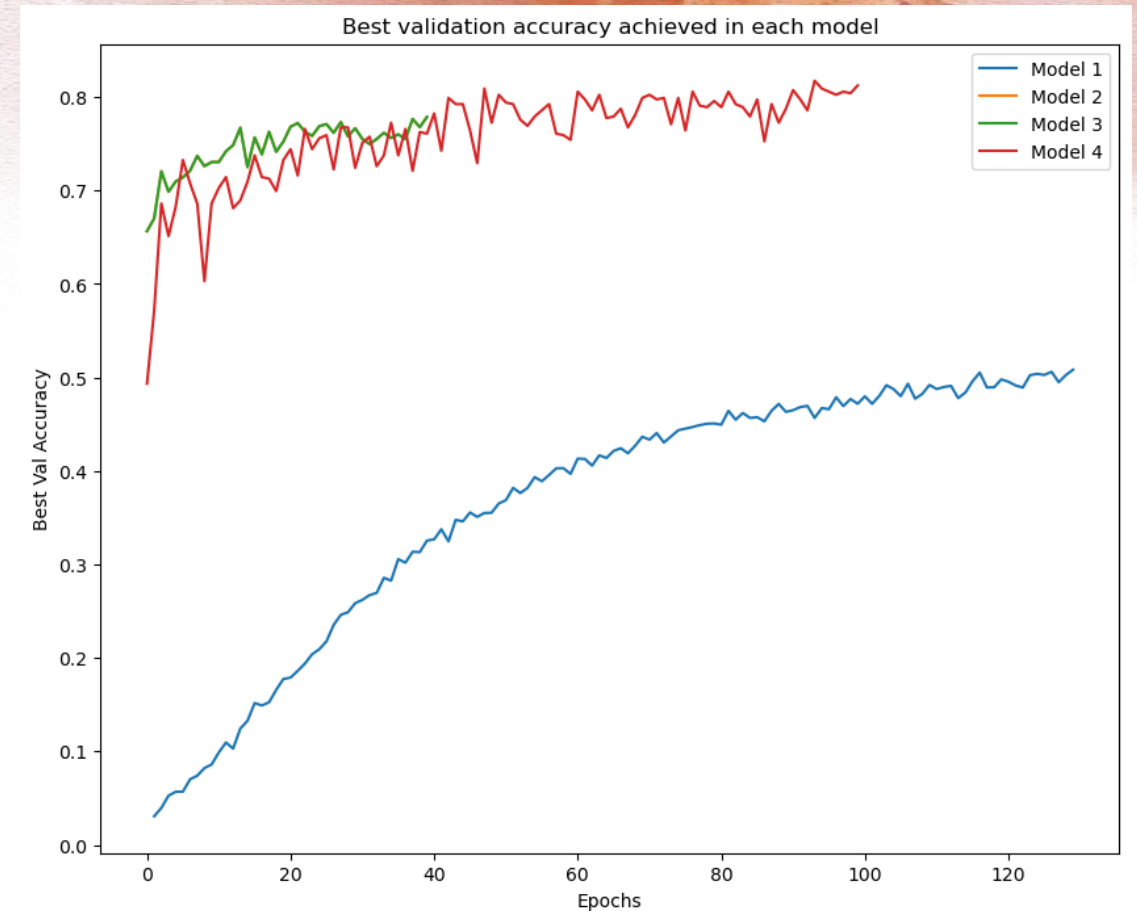


Fig: Validation Accuracy Trend Across Epochs for Multiple Models

Model Testing Using Web Integration

Implemented Specification:

Framework Used: Stream-lit

Tools Used : Python

Model Chosen :

Based on the training results shown from the previous slides the models are selected (Dense-Net)

Note:

- As of now, Dense-Net is proven to be giving maximum accuracy among the other trained models.
- This model can be further enhanced or improved based on the continuous training of the model.
- Web Integration done using Stream-lit is carried out on local machine.

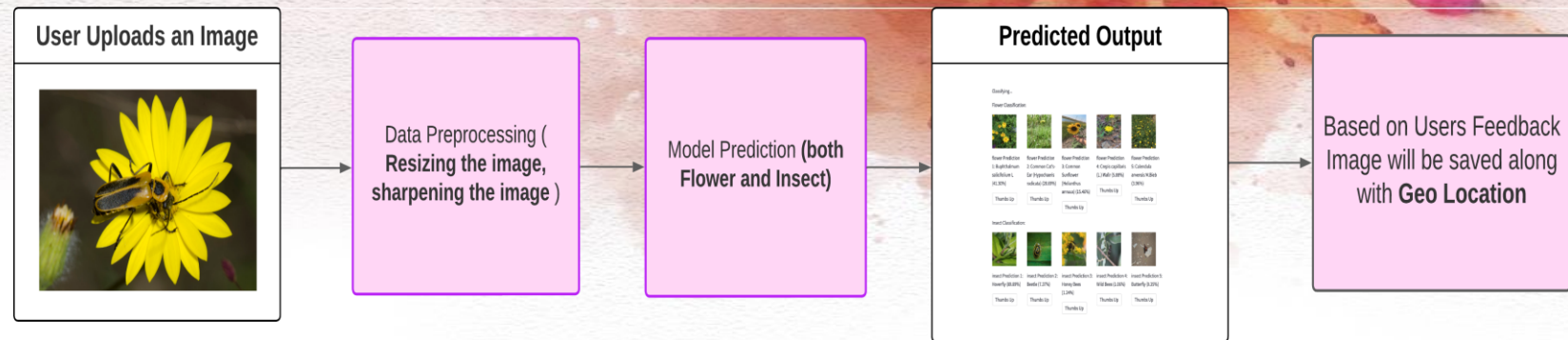


Fig: Web Integration Prototype

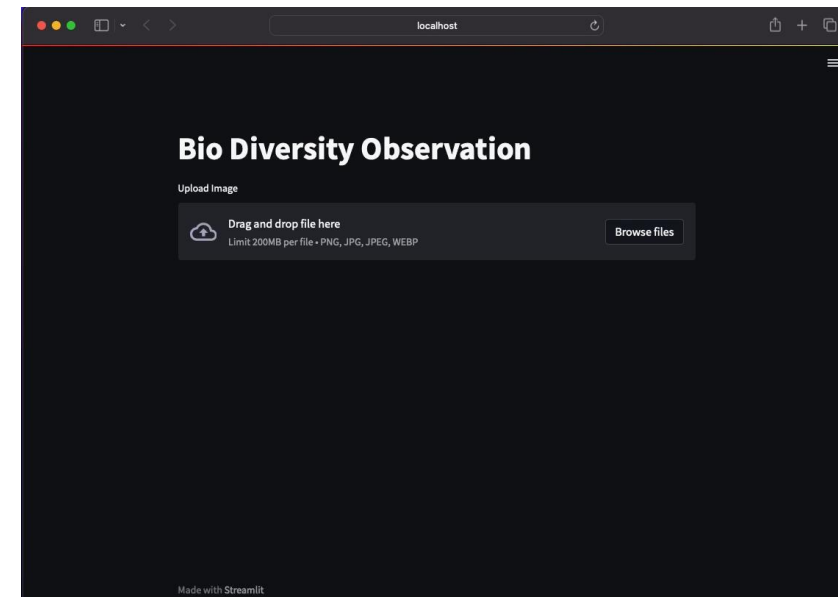
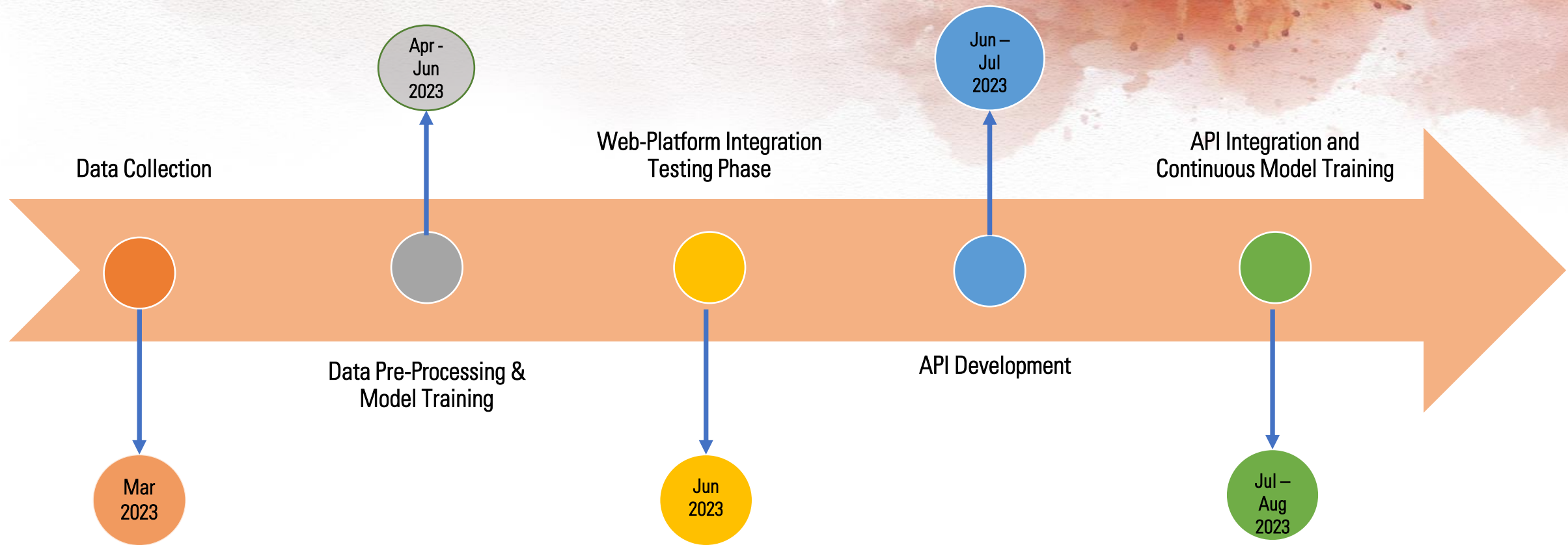


Fig: Prototype Demo

Milestone



THANK YOU